

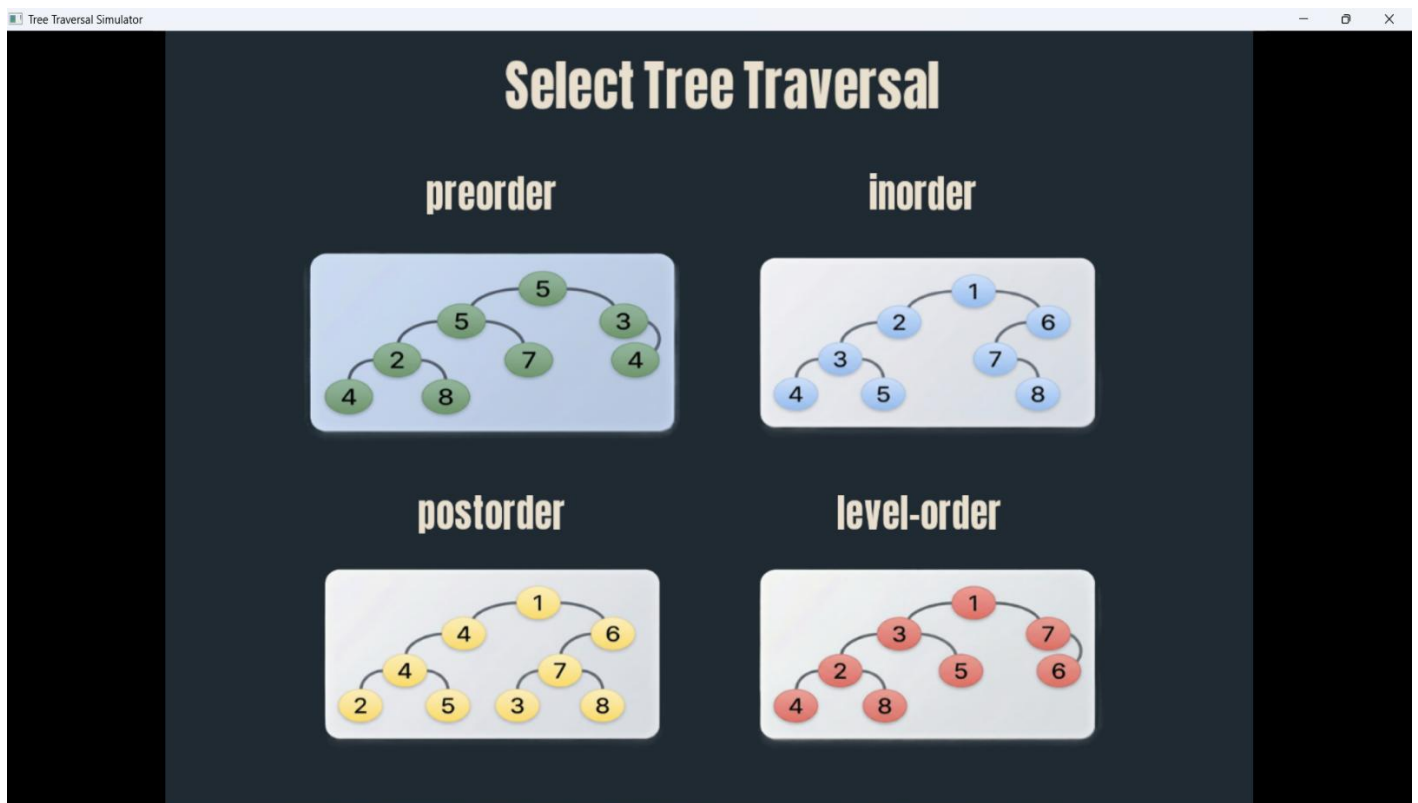
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ATHENS UNIVERSITY OF ECONOMICS AND BUSINESS
Assignment in Computer Programming with C++

The application's User Guide

**INTERACTIVE APPLICATION USING A NODE NETWORK:
REPRESENTATION OF TREE TRAVERSALS IN A USER-CREATED
BINARY TREE**

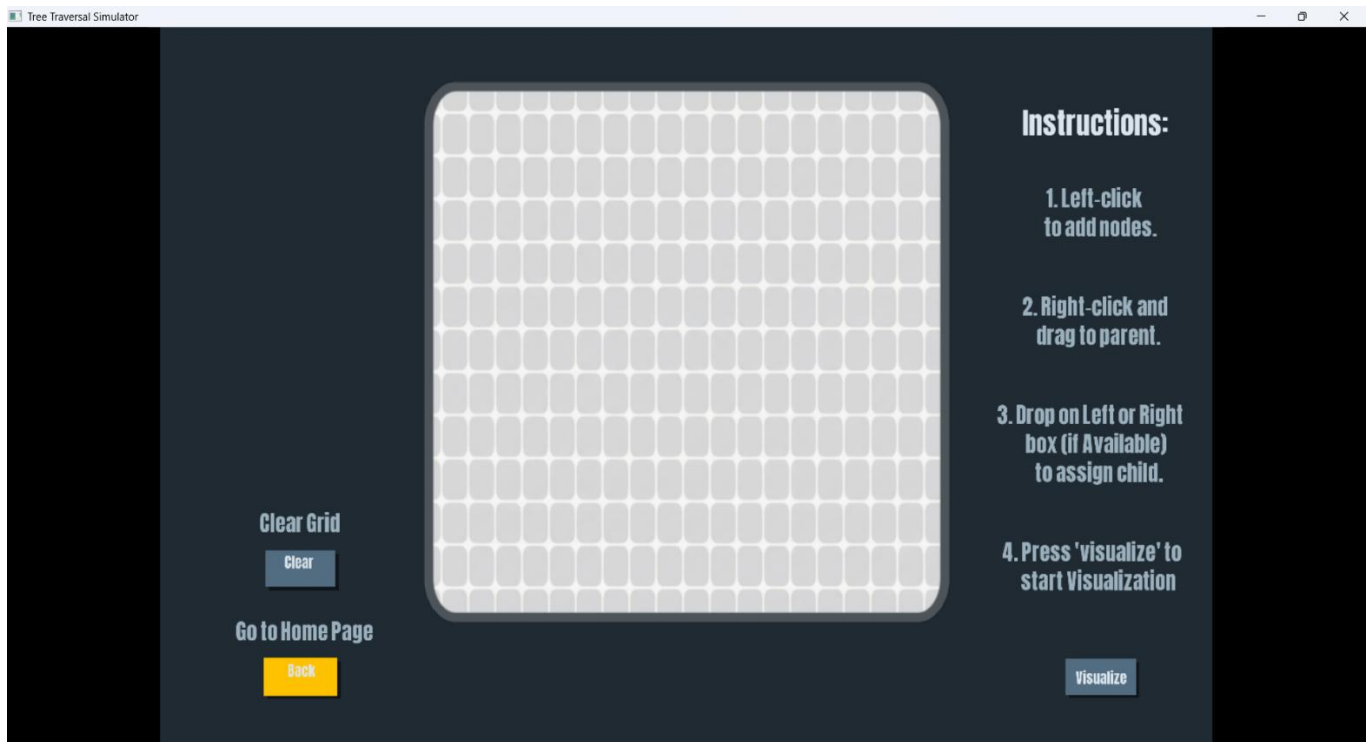
APPLICATION EXECUTION OVERVIEW (RUN-THROUGH)

1. HOME SCREEN – Start Level class (Level 1)



 Select the traversal method for the binary tree you will build next by left-clicking on one of the 4 buttons with your mouse.

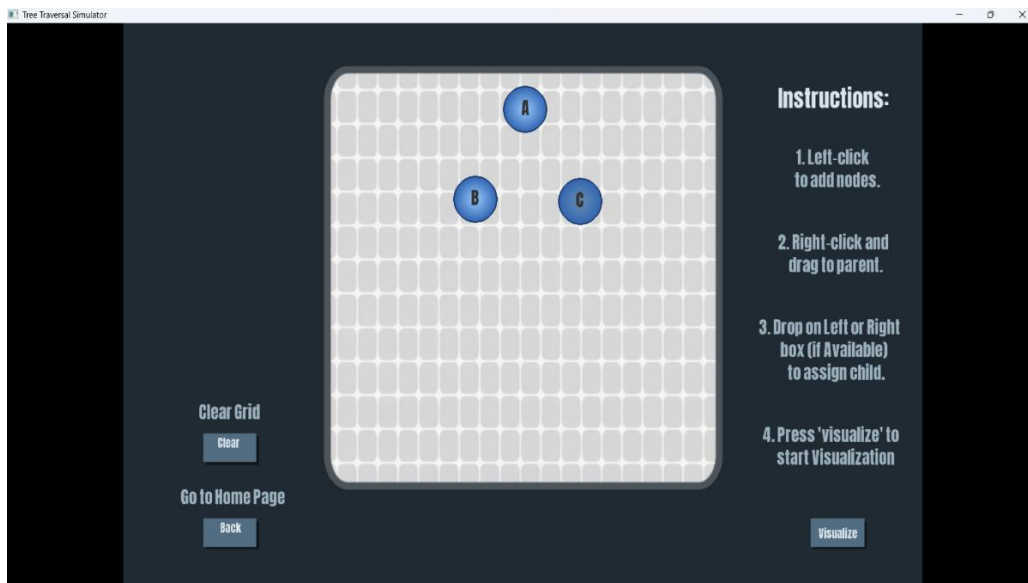
2. EDITING LEVEL (Level 2)



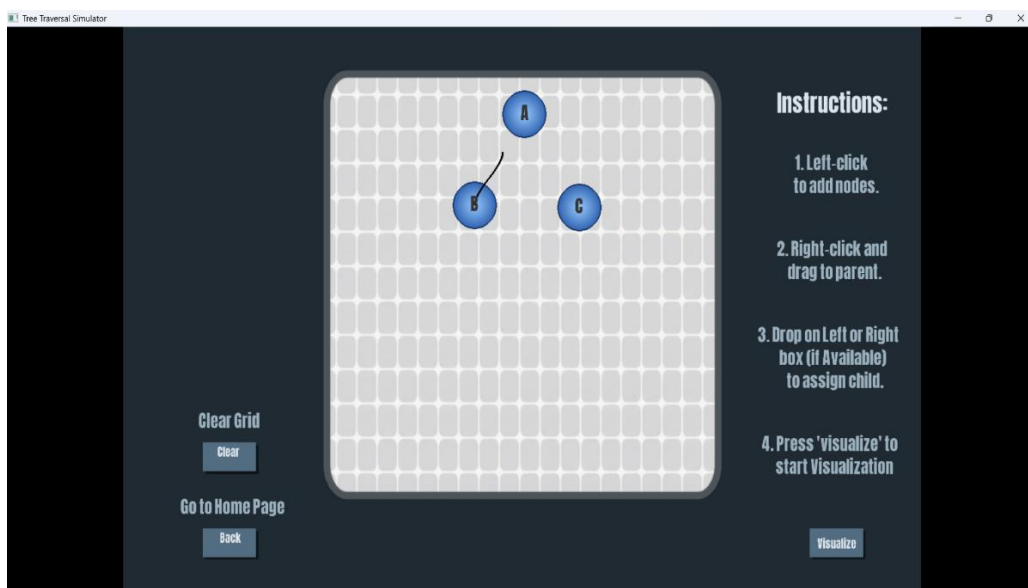
- ✚ This is the level where the user creates the binary tree, strictly within the boundaries of the square grid at the center of the screen, as generating tree nodes and edges outside these limits is restricted.
 - ✚ Initially, the grid is empty.
 - ✚ On the left of the grid, we have:
 - The Clear Grid button: When clicked (mouse left-click), it deletes all nodes and edges generated by the user so far, thereby resetting/clearing the grid.
 - The Back button: When clicked (mouse left-click), it returns the user to the Home Screen (Level 1).
 - ✚ On the right of the grid, we have:
 - User Instructions (Manual): Step-by-step guidance for building the tree (Steps 1. to 4.).
 - The Visualize button: When clicked (mouse left-click), the binary tree traversal simulation begins, transitioning to the Visualization Level (Level 3).
- ✚ Furthermore:

BINARY TREE CONSTRUCTION TUTORIAL

- ✓ **STEP 1 - NODES GENERATION**: Left-click anywhere within the boundaries of the grid to generate the nodes of the tree.

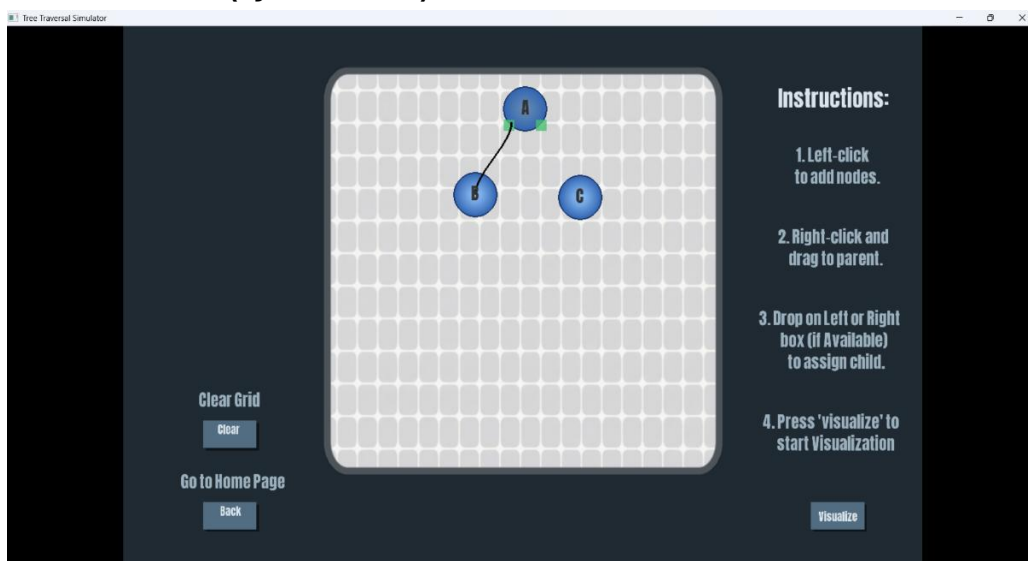


- ✓ **STEP 2 – CONNECTING TWO NODES (GENERATING AN EDGE BETWEEN THE CHILD-NODE AND THE PARENT-NODE)**:
- Right-click (mouse) over the node intended to be the child node between the two (in this case, node B).
 - While holding down the right mouse button, drag toward the node intended to be the parent node (in this case, node A). In this way, you can generate an edge between the two nodes, in order to connect them.

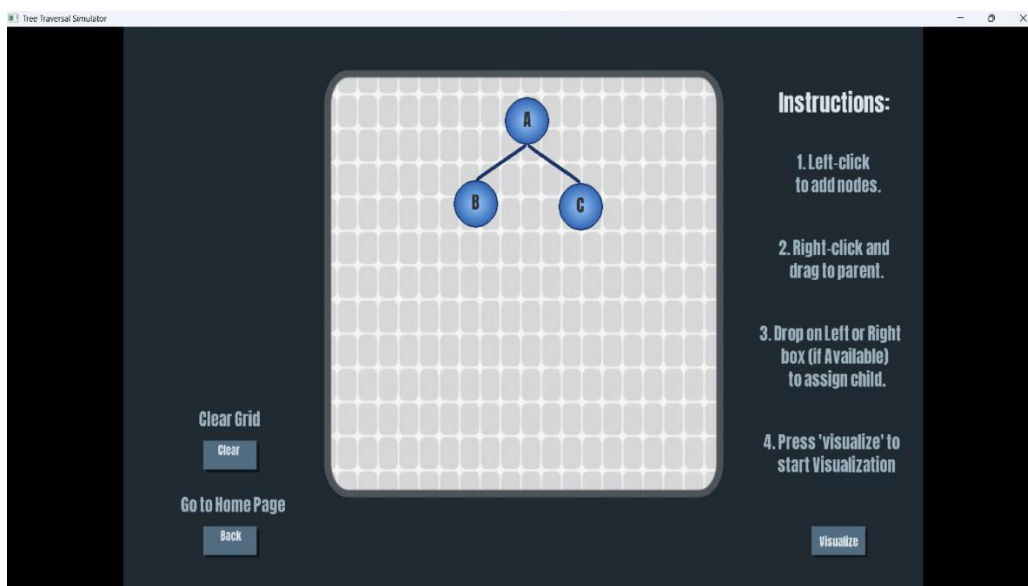


✓ **STEP 3 – ASSIGN THE CHILD-NODE AS A LEFT-CHILD OR A RIGHT-CHILD TO THE PARENT-NODE.**

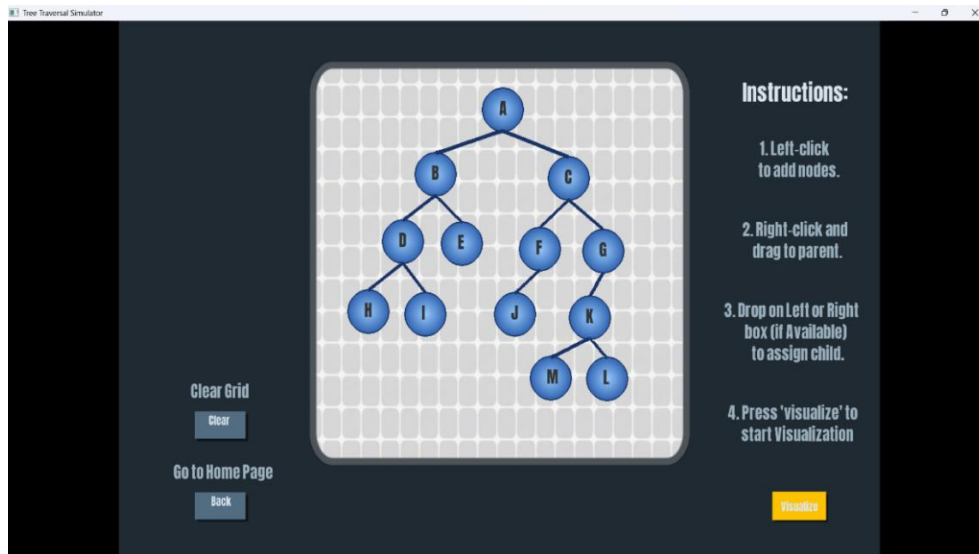
- Keep dragging your mouse (second-end of the edge connecting the nodes) toward the parent-node to make the 2 small green “connecting squares” pop-up.
- Drop the end of the edge (by stop dragging your mouse) on the left green square (same as the following pic) to assign the child-node (node B) as a left-child to the parent-node (node A), or:
- Otherwise, assign the child-node as a right child (node C) by dropping the end of the edge on the right green square, the same way as described above (symmetrical).



▪ **Result:**



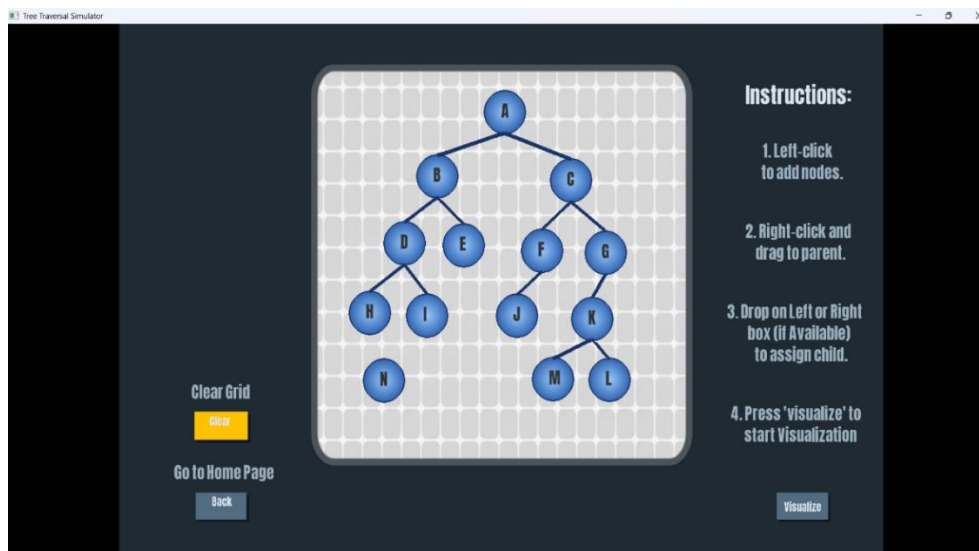
- ✓ Example of a binary tree constructed according to the methodology described above:



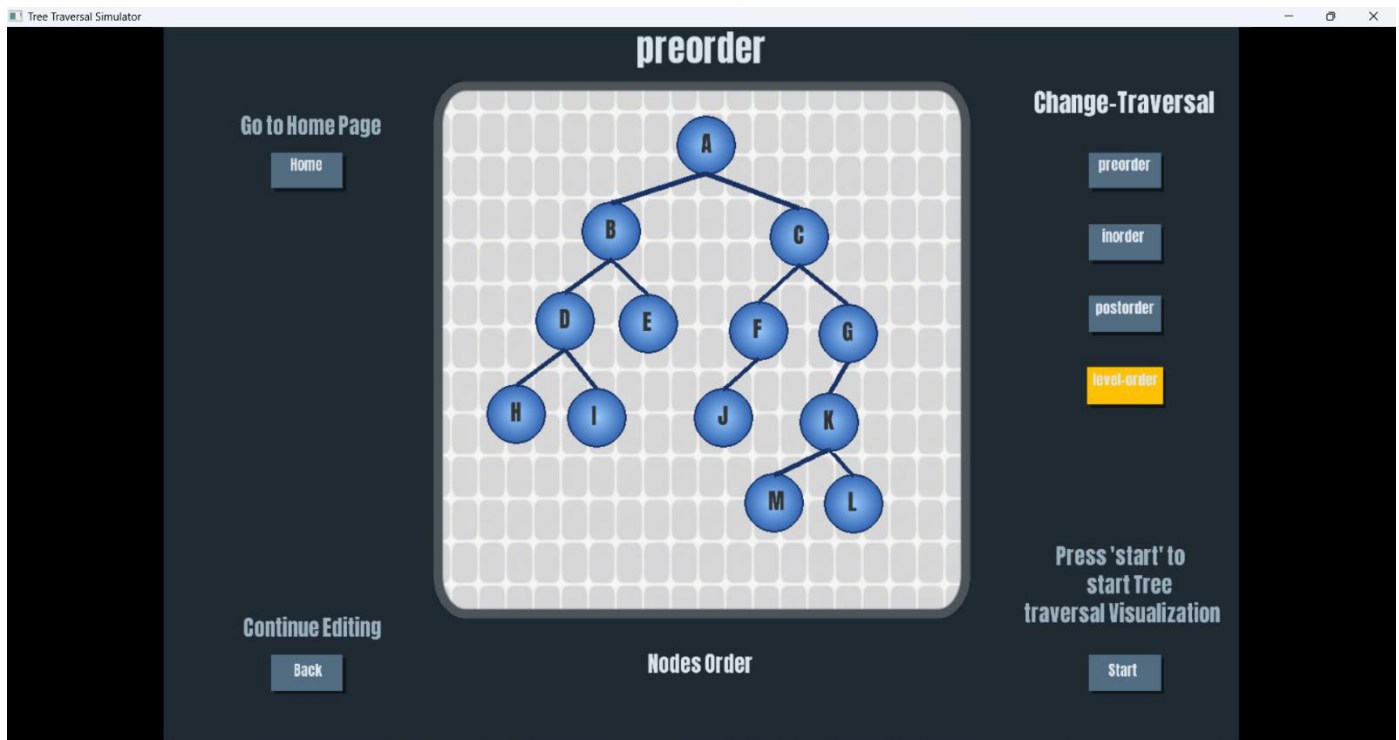
✓ **STEP 4 – TRANSITION TO THE VISUALIZATION-LEVEL (Level 3):**

Left-click with your mouse, on the bottom right button, to move on to the next level, after constructing your binary tree.

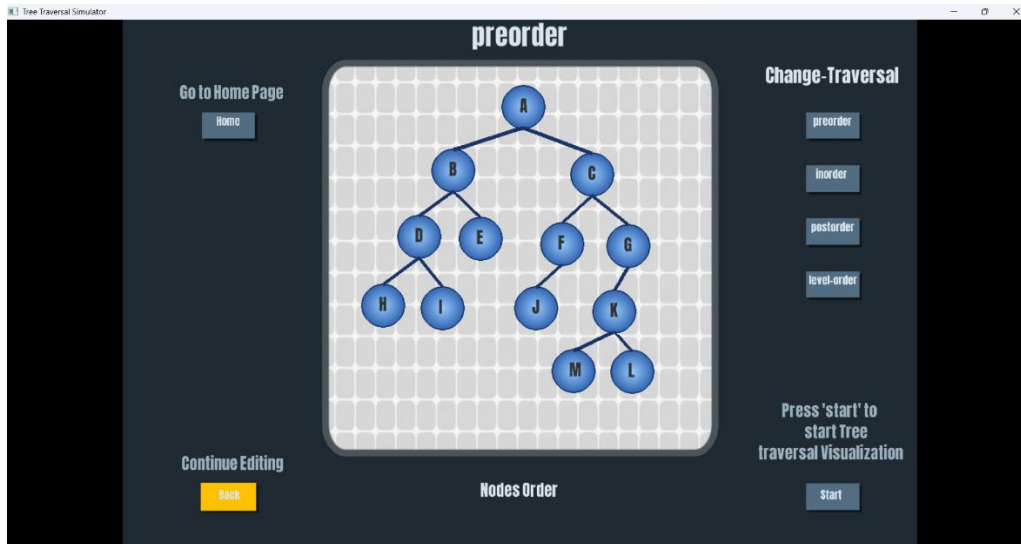
- **NOTE:** Any isolated nodes not connected to the rest of the tree (leftovers) are not considered part of the tree and will automatically be deleted from the grid upon transitioning to Level 3. (for example, Node N in the following pic)



3. VISUALIZATION LEVEL (Level 3)



- ✚ This is the level where the user watches the tree traversal through a slow-paced, stop-motion node color-changing animation (described below), according to the tree traversal selection made initially.
- ✚ At this stage, the tree can no longer be modified as in the Editing Level (Level 2) and remains static (fixed). Any leftover nodes have been removed (as described previously).
- ✚ At the top of the screen, above the grid, we have:
 - A label indicating the binary tree traversal method currently selected by the user (e.g., "preorder" in the image above).
- ✚ On the left of the grid, we have:
 - The Go to Home Page button: When clicked (mouse left-click), it returns the user to the initial Home Screen (Level 1), restarting the entire interactive educational application from the beginning.
 - The Back button ("Continue Editing"): When clicked (mouse left-click), it returns the user to the Editing Level (Level 2), allowing them to continue modifying their tree (the exact same tree previously built, unchanged) and add new nodes and edges.



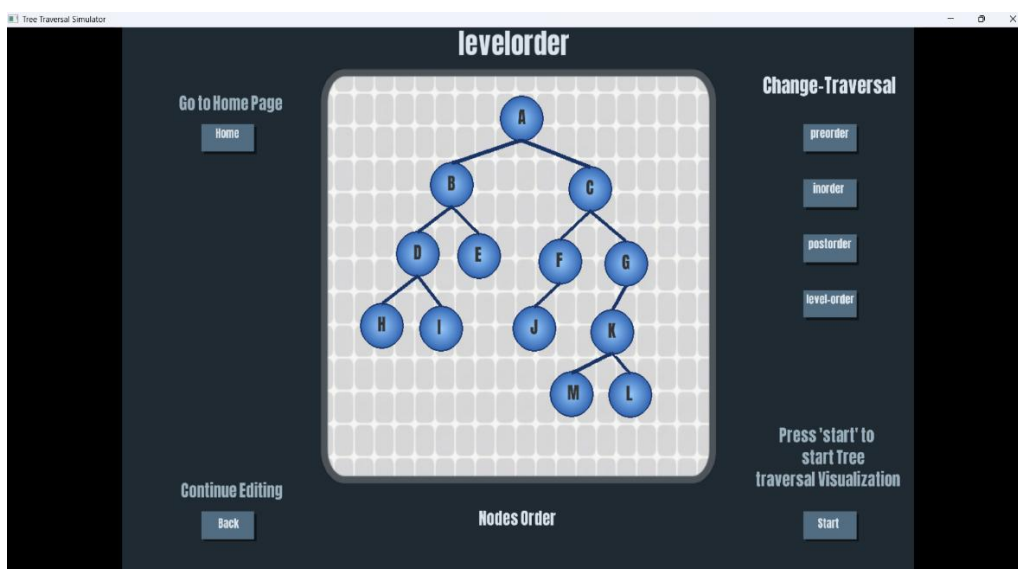
On the right of the grid, we have:

- The four Change-Traversal buttons: One button for each traversal method (preorder, inorder, postorder, levelorder). When clicked (mouse left-click), they allow the user to switch the traversal method or try out any of the remaining methods after their initial selection.

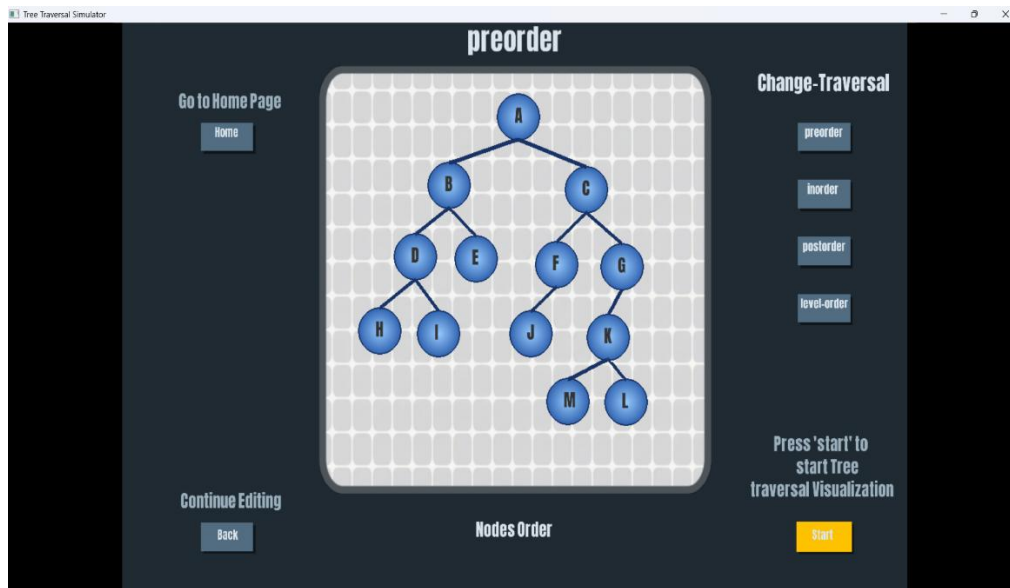
Furthermore, switching the traversal method in this way dynamically updates the label at the top of the screen above the grid.

Example: Switching from preorder (initial choice) to level-order:

(As shown two images above, hovering the mouse over the button changes its color to yellow, followed by a mouse left-click on the "level-order" button).

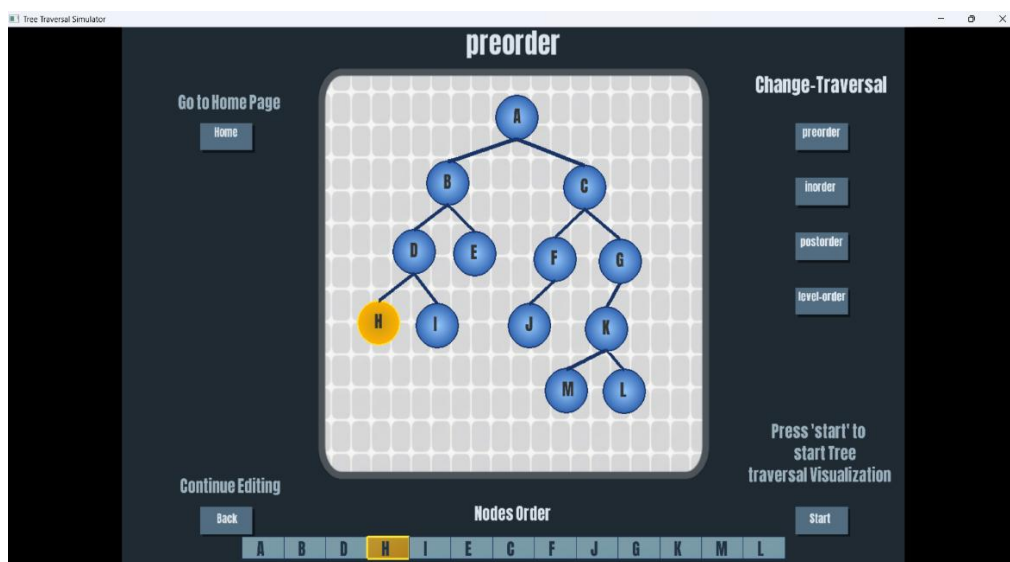


- The **Start button** (“Press ‘start’ to start Tree traversal Visualization”): When clicked (mouse left-click), it initiates a slow stop-motion node color-changing animation (described below) according to the selected traversal method by the user.



When the “Start” button is left-clicked, on the bottom of the screen, under the grid:

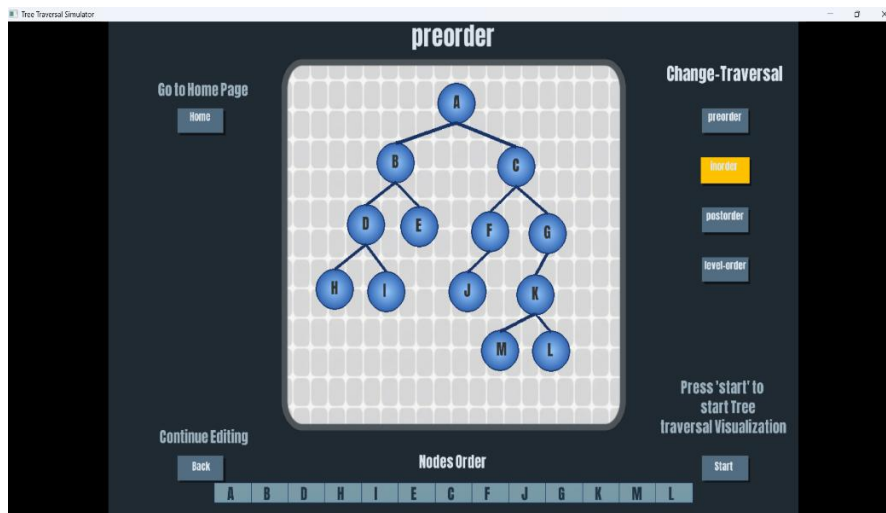
- A queue appears (“Nodes Order”), displaying the correct sequence of nodes (each box in the queue represents a node's letter, as shown below) based on the selected traversal method. Simultaneously, during the animation referenced above, the tree nodes turn yellow one by one with a step, in the order of the selected traversal, while the corresponding box containing the node's letter in the queue also turns yellow (e.g., node H in the image).



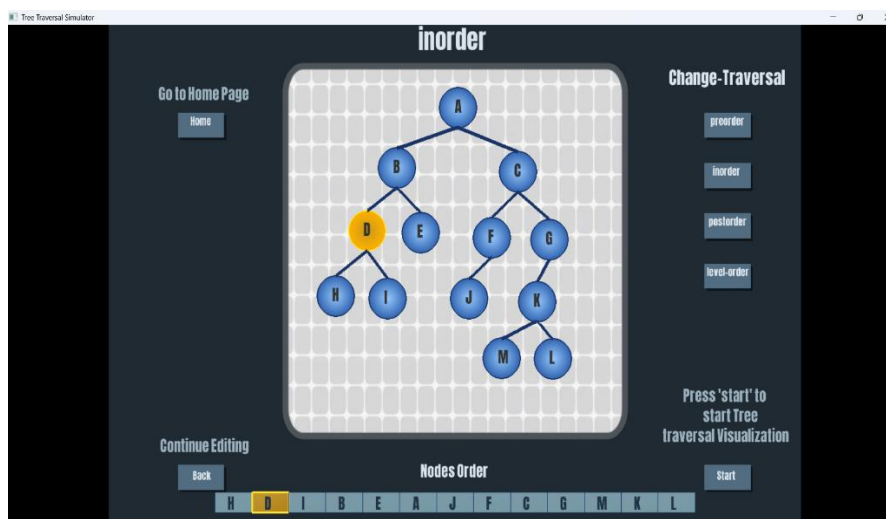
Other features and examples

🌲 Change from “preorder” traversal to “inorder” traversal in the Visualization Level (Level 3):

- Mouse left-click on the “inorder” button on the right of the grid (the button changes to a yellow color when you hover above it)

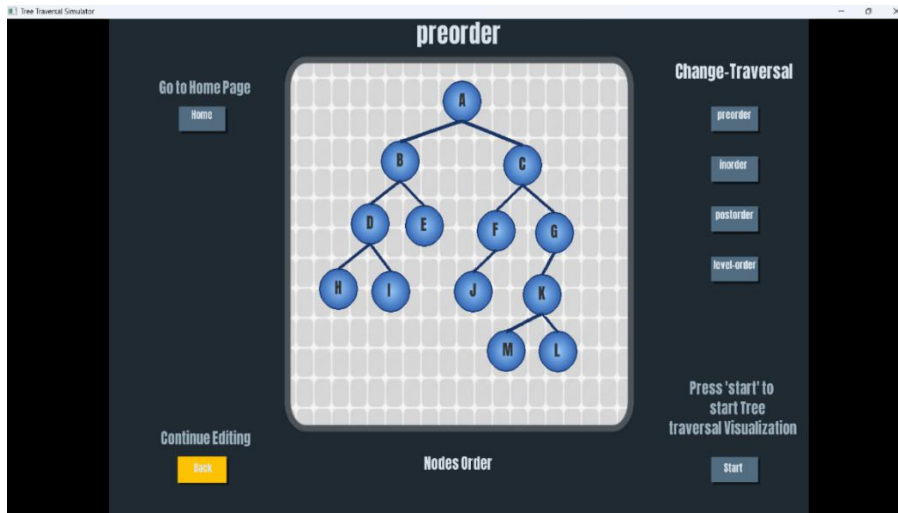


→ Press ‘start’ to start Tree Traversal Visualization (bottom right)



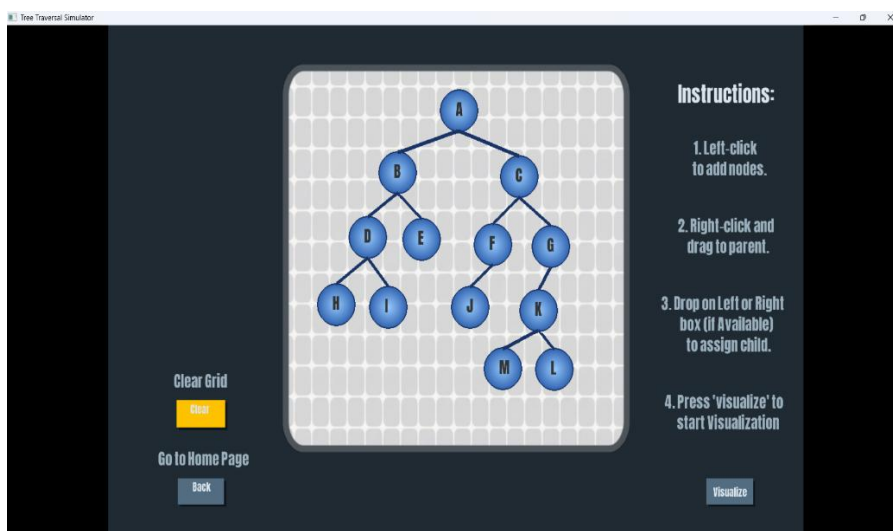
- Change from the Visualization Level (Level 3) to the Editing Level (Level 2), by mouse left-clicking the “Back” Button (bottom left corner, “Continue Editing” Label)

- Level 3 (Visualization)

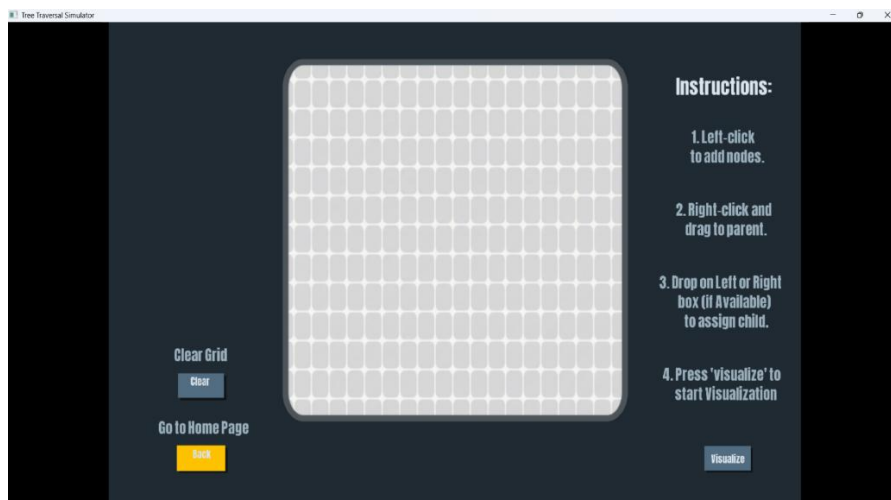


- Level 2 (Editing)

- Clear the Grid (that is deleting the whole tree) by left-clicking with your mouse on the “Clear” button (bottom left corner). Next, we can create a new binary tree, starting from the beginning all over again!

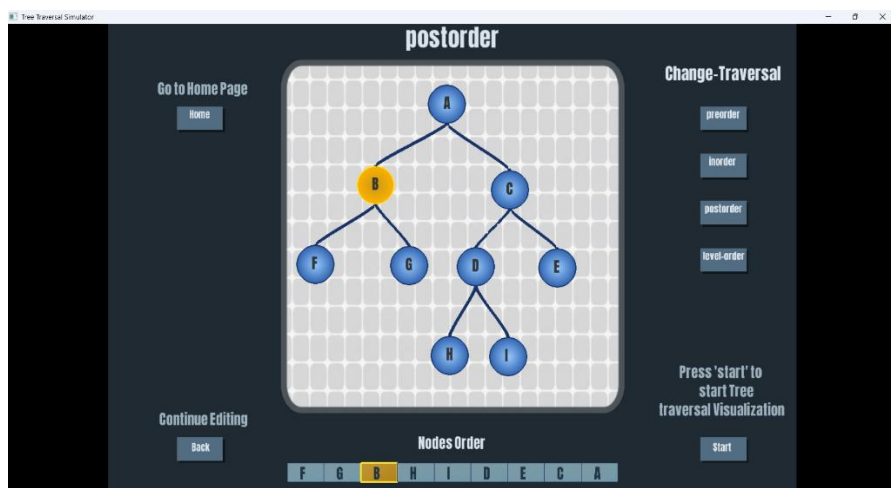


🌈 A new, clean and empty grid:

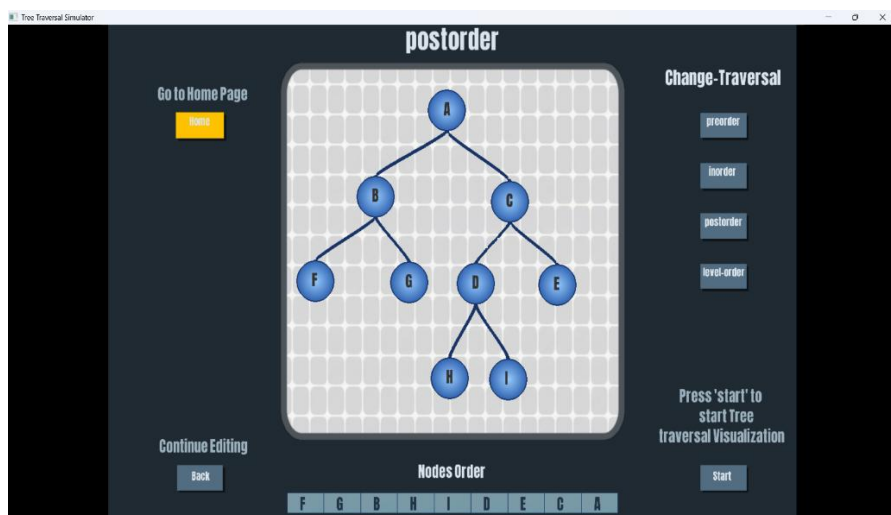


➔ A new binary tree created:

🌈 “Postorder” traversal method selected, a screenshot from the visualization animation process in Level 3:



🌈 Return to Level 1, by left-clicking the “Go to Home Page” button (top left corner)



→ Return to the Home Page (Level 1):

